

SOP #16 – ATI (Analysis to Improve) for Fire Suppression Systems

Goal: Maintain full compliance and operational readiness while reducing false alarms and minimizing downtime

1. Performance Metrics

Metric	Benchmark	Strategy for Improvement
System Response Time	<60 seconds	Routine alarm and flow testing
Preventive Maintenance Compliance	100% completion monthly	Calendar tracking and visual logs
Incident Response Rate	<5% of activations	Improve staff training and signage
Alarm Accuracy	≥95% valid activations	Calibrate sensors quarterly

2. Optimization Strategies

- Install isolation valves per zone to limit service interruptions
- Digitally monitor pressure zones via smart gauges or BMS
- Add desiccant dryers in dry systems to reduce corrosion
- Upgrade panel communication to IP-based systems for alerts
- Conduct biannual EHS evacuation drills with log reviews

3. Legal & Safety Compliance

- NFPA 13, 25, and 72 – Sprinkler design, inspection, testing
- OSHA 1910 Subpart L – Fire protection and life safety requirements
- AHJ Reporting Requirements – Maintain logs for at least 3 years
- FM Global / ISO Ratings – Risk score improvements via upgrades
- Insurance Policy Conditions – May require third-party inspection